

Exercice : Diagramme $T_i - V$

Q1. A: liquide $T_i + V$ homogène $\varphi = 1$
B+C: liquide $T_i + V$ homogène + solution solide
 $T_i - V$ $\varphi = 2$

Q2. Point I $\equiv$ point invariant
Il s'agit d'une composition pour laquelle la solution solide fond de manière congruente à température constante.

Par conséquent, en ce point, $w_V^P(T_I) = w_V^S(T_I)$

Q3. Curves de refroidissement: isobares



Calcul de la variance particulière:

A-B: X: $P, T, w_V^P, w_{T_i}^P$
Y: $w_V^P + w_{T_i}^P = 1$
Z: $P = \text{cte}$ $V_p = 2$

B-C: X: $P, T, w_V^P, w_{T_i}^P, w_V^S, w_{T_i}^S$
Y: $w_V^P + w_{T_i}^P = 1$
 $w_V^S + w_{T_i}^S = 1$
 $w_V^P(T_i) = w_V^S(T_i)$

$$\begin{aligned} \mu_V^P &= \mu_V^S \\ \mu_{Ti}^P &= \mu_{Ti}^S \\ Z: P &= \text{cte} \end{aligned}$$

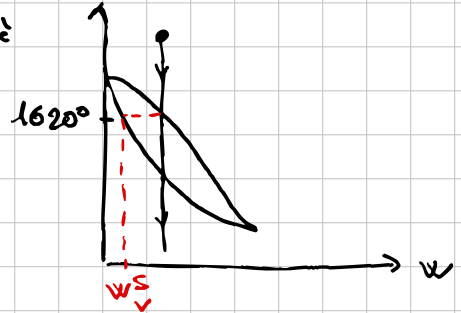
$$v_p = 6 - 5 - 1 = 0$$

C-D: X: T, P, w_V^S, w_{Ti}^S
 Y: $w_V^S + w_{Ti}^S = 1$
 Z: $P = \text{cte}$

$$v_p = 4 - 1 - 1 = 2$$

Q4. le premier cristal apparaît à 1620° et de composition en vanadium:

$$w_V^S(1620^\circ) = 0.04$$



Q5. Calcul de la fraction molaire en V:

$$w_V = \frac{m_V}{m_V + m_{Ti}} = 0.1$$

Composition:
$$\begin{cases} w_V^S(1600^\circ) = 0.07 \\ w_V^P(1600^\circ) = 0.17 \end{cases}$$

Théorème de lever des bras:

$$w^P = \frac{w_V - w_V^S}{w_V^P - w_V^S} \stackrel{\text{A.W}}{=} \frac{0.1 - 0.07}{0.17 - 0.07} = 0.3$$

$$w^S = 1 - w^P = 0.7$$

À 1600°C, le mélange contient 300 kg de liquide et 700 kg de solide.

Calcul de la composition:

$$m_V^P = w_V^P m^P$$

$$m_{T.i.}^P = (1 - w_V^P) m^P$$

$$m_V^S = w_V^S m^S$$

$$m_{T.i.}^S = (1 - w_V^S) m^S$$

A.N.:

$m_V^S = 49 \text{ kg}$	$m_{T.i.}^S = 651 \text{ kg}$
$m_V^P = 51 \text{ kg}$	$m_{T.i.}^P = 249 \text{ kg}$